Disc Harrow Metric Calculation & Sizing Guide

Pure Metric Formulas, Disc Count Calculations & Power Requirements

1. DISC COUNT & SPACING FORMULA

Calculating Total Blade Count on a Harrow

To determine total disc count based on working width, blade spacing, and gang arrangement:

```
Total Discs = [ Working Width (m) / Blade Spacing (m) ] + Number of Gangs
```

2. DISC COUNT QUICK REFERENCE GUIDE

Harrow Category	Working Width (m)	Blade Spacing (mm)	Typical Disc Count
Light / Orchard	1.5 – 2.0 m	180 – 200 mm	12 – 16 Discs
Medium Tandem	2.5 – 3.5 m	200 – 230 mm	20 – 32 Discs
Heavy Offset	3.5 – 5.0 m	230 – 280 mm	32 – 44 Discs
Broadacre Folding	6.0 – 9.0+ m	230 – 280 mm	52 – 80+ Discs

3. SOIL DRAFT FORCE REFERENCE (KN/M)

Specific draft resistance per meter of working width across various depths:

Soil Type	Light (50–80 mm)	Medium (80–120 mm)	Heavy (120–150 mm)
Sandy Soil	1.5 – 2.5 kN/m	2.5 – 4.0 kN/m	4.0 – 5.5 kN/m
Loam Soil	2.5 – 4.0 kN/m	4.0 – 6.0 kN/m	6.0 – 8.5 kN/m
Heavy Clay	4.0 – 6.0 kN/m	6.0 – 9.0 kN/m	9.0 – 12.0 kN/m

4. DRAWBAR POWER & ENGINE KW CALCULATION

Metric Power Formulas

Convert draft force and field operating speed into required engine power:

```
Drawbar Power (kW) = [ Total Draft Force (kN) × Speed (km/h) ] / 3.6
```

```
Engine Power Needed (kW) = Drawbar Power (kW) / 0.75
```

Note: Required engine horsepower = Engine kW × 1.36.

5. PRACTICAL CALCULATION EXAMPLE

Example: Sizing a 3.0 m Medium Tandem Disc Harrow

- **Implement Specs:** 3.0 m width, 230 mm blade spacing, 4 gangs = **28 total discs**
- **Operating Parameters:** Medium loam soil, 100 mm depth (≈ 5.0 kN/m draft), speed = 8 km/h
- **Step 1 (Total Draft):** 3.0 m × 5.0 kN/m = 15.0 kN draft force
- **Step 2 (Drawbar Power):** (15.0 kN × 8 km/h) / 3.6 = 33.3 kW drawbar power
- **Step 3 (Engine Power):** 33.3 kW / 0.75 = **44.4 kW Engine Power (approx. 60 HP)**
- **Step 4 (Power per Disc):** 44.4 kW / 28 discs = **1.58 kW per disc**

6. TRACTOR POWER VS. DISC COUNT QUICK REFERENCE

Tractor Power (kW / HP)	Recommended Disc Count	Working Width (Medium Soil)	Target Speed Range
35 – 55 kW (48–75 HP)	16 – 24 Discs	2.0 – 2.8 m	7.0 – 9.0 km/h
60 – 85 kW (80–115 HP)	28 – 36 Discs	3.0 – 4.2 m	7.5 – 9.5 km/h
90 – 130 kW (122–175 HP)	40 – 52 Discs	4.5 – 6.0 m	8.0 – 10.0 km/h
140+ kW (190+ HP)	56+ Discs	6.5+ m	8.5 – 10.5 km/h